Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - II

Dewatering and Grouting

Syllabus topics covered in this unit

- Methods of De-watering
- Sumps and Interceptor Ditches
- Single and Multi Stage Well Points, Vacuum Well Points
- Horizontal Wells, Foundation Drains, Blanket Drains
- Criteria for Selection of Fill Material Around Drains
- Electro-osmosis
- Objectives of Grouting, Grouts and Their Properties
- Grouting Methods: Ascending, Descending and Stage Grouting
- Hydraulic Fracturing in Soils and Rocks
- Post Grout Tests

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Dewatering - Introduction and Methods

Dewatering is the removal of ground water or surface water from a construction site (excavation, trench, foundation, tunnel) to keep the work area dry and the sides stable. It is also used to lower the water table permanently (drains) to improve the soil strength.

Need for dewatering

- Excavation below the water table is unsafe - sides collapse, bottom heaves, quick condition (piping) develops when the upward seepage gradient $i = i_c = (G - 1)/(1 + e)$ about 1.
- Dry condition is needed for foundation work, concreting and compaction; reduce the lateral pressure on the excavation supports; increase the effective stress and thus the strength; avoid buoyancy of structures.

Classification of methods

Group	Methods	Suitable soil
1. Open sump / ditch (surface) methods	Sumps, interceptor ditches, pumping from open pits	Coarse soils: gravel, coarse sand, rock; clean low flow
2. Well point systems	Single stage, multi stage, vacuum, eductor, deep well	Sand, silty sand (k about 10^{-2} to 10^{-5} cm/s)
3. Deep wells	Deep well with submersible pump, ejector wells	Large drawdown > 15 m, permeable soils
4. Horizontal wells / drains	Horizontal drains, foundation drains, blanket drains	Slopes, dams, long excavations
5. Electro-osmosis	Direct current across soil	Silts, clays (k < 10^{-5} cm/s)
6. Cut-off and exclusion methods	Sheet piles, diaphragm wall, slurry trench, grouting, freezing	Where pumping is not allowed or inflow is large

The choice depends on permeability (grain size), depth of the draw down, area to be dewatered, duration, cost, effect on neighbours. A general guide is: gravel - sump; sand - well points; fine sand and silt - vacuum well point; clay - electro-osmosis.

Darcy and flow to a well

Basic formulae

- Darcy: $q = k\,i\,A$. Steady flow to a fully penetrating well in an unconfined aquifer (Dupuit): $Q = \pi k (H^2 - h^2) / \ln (R/r)$
- Confined aquifer (thickness b): $Q = 2\pi k b (H - h) / \ln (R/r)$
- R = radius of influence (Sichardt): $R = 3000\, s \sqrt{k}$ (s = drawdown in m, k in m/s, R in m)
- For a long trench / line of wells: $q = k (H^2 - h^2) / (2L)$ per unit length.

2. Sumps and Interceptor Ditches

Sump pumping (open pumping)

A sump is a pit dug lower than the excavation level in a corner or along the side, at which water seeps in by gravity and is pumped out by a centrifugal / submersible pump. It is the simplest and cheapest method.

- Sump size: about 1 m x 1 m or more, depth 1 - 2 m below the formation level; lined with perforated drum / precast rings / brick with gravel filter to prevent fines from entering.
- Sump pit at the lowest corner; a perimeter drain leads water to the sump. Pump capacity > the inflow.
- Advantages: simple, cheap, small plant, suited to gravel and coarse sand and rock excavation with small depth (< 1.5 - 2 m below water table).
- Limitations: in fine sands the upward seepage causes piping / quick sand, boiling at the bottom, sloughing of slopes, unstable; depth limited by suction lift (about 6 - 7 m max).

Interceptor (cut-off) ditches

- Interceptor ditch (catch drain) is a ditch dug around the upslope side of the excavation or slope to intercept the surface runoff and shallow ground water before it reaches the works. The water is led by gravity to a natural drain or to a sump.
- Section: trapezoidal 0.5 - 1.0 m wide; the invert slope 1 in 200 or more; sometimes lined with concrete or geotextile and filled with gravel (French drain) with perforated pipe.
- Used for road cuttings, hill-side slopes, embankment toe, dam abutments, landslide stabilisation; lowering the water table in the slope improves stability.
- Interceptor drains are also built as a trench perpendicular to the flow of the groundwater, in which a perforated pipe in filter material collects the water.

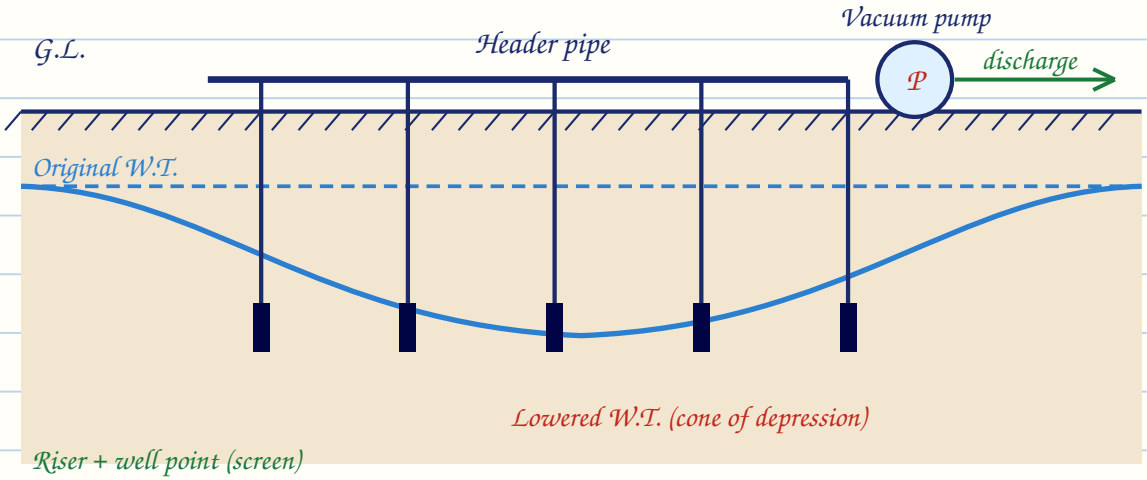
Surface water and seepage → Interceptor ditch (collect) → Sump → Pump and discharge

Remember

- Sump and ditch methods = open pumping, lowest cost, but not for fine soils or deep excavation.
- Filter around the sump pit is a must; slope the excavation floor towards the sump.

3. Well Point Systems

A well point is a small diameter (40 - 50 mm) perforated tube (screen about 0.5 - 1 m long and covered with a mesh) driven or jetted into the ground. A line of well points spaced 0.75 - 3 m apart (usually 1 - 2 m) is connected to a common header pipe (150 - 200 mm) and a centrifugal / vacuum pump that draws the water by suction.



Components

- Well point (screen + tip with jetting valve), riser pipe (40 - 50 mm), swing connection (flexible hose), header pipe, pump (vacuum assisted self-priming) and the discharge pipe.
- Installation: by jetting with water pressure (6 - 8 kg/cm^2) into the sand, giving a filter sand column of coarse sand around the screen, in a hole of 20 - 25 cm.

Single stage well points

- One line (or ring) of well points around the excavation; the water table is lowered up to about 5 - 6 m below the pump level (limit of the suction lift).

- The header at about the ground level. Spacing: 1 m in fine sand, 2 m in coarse sand; use closer spacing at the more permeable parts. A double sided system (both sides of the trench) for wide excavation > 6 m wide.
- Pump capacity: yield of one well point 0.1 - 1 l/s; a pump of 150 mm handles 100 - 150 well points.

Multi stage well points

- For depth of the draw down greater than about 5 - 6 m: well point lines are set at two or more levels (stages) one above the other in a stepped excavation (benches) - the upper stage is installed first and excavated, then the next stage lower.
- Each stage lowers the water table by 4 - 5 m. Needs a wide excavation (benches 1.5 m or more wide, side slopes), more pumps, more cost, large water volume. Alternative: deep wells / ejector well points if the depth > 12 m.

Vacuum well points

- In silty and fine sands ($k < 10^{-3}$ cm/s), gravity flow is slow, so a vacuum (about 0.6 - 0.8 atmosphere) is maintained in the well point and sand filter (by sealing the top of the filter with clay/ bentonite). The vacuum increases the hydraulic gradient and the soil water is drawn out; it also consolidates the soil (additional effective stress).
- The filter sand column up to the ground surface is sealed. Spacing is closer (0.75 - 1.5 m). Used for deep trenches in fine silty soils, stabilising slopes, tunnels in silty sand.
- Ejector (eductor) wells use high pressure water jet (ejector) at the bottom - for depth up to 30 m at low yield in fine soils.

Aspect	Single stage	Multi stage	Vacuum
Max drawdown	About 5 - 6 m	10 - 15 m (2 or 3 stages)	Increases gradient; about 6 m per stage
Soil	Sand, coarse silt	Sand	Silty sand, fine silt
Spacing	1 - 3 m	1 - 2 m per stage	0.75 - 1.5 m
Cost	Low	High	Medium - High

Design of a well point system (steps)

Soil data and k → Required drawdown, flow Q → No. of well points = Q / q_{wp} → Pump and header selection

- Total flow to the system is found by treating the system as a large well of equivalent radius r_e = sqrt(A / pi) (A = area enclosed): $Q = \pi k (H^2 - h^2) / \ln (R / r_e)$ for an unconfined aquifer.
- The drawdown at the centre is s, and extra 0.5 - 1 m below the formation level is provided. Well point yield $q_{wp} = 2 \pi r_s L_s$ (sqrt(k) / 15) (Kozeny / Sichardt).

4. Horizontal Wells, Foundation Drains and Blanket Drains

Horizontal wells / horizontal drains

- Horizontal (or slightly upward inclined) perforated pipes (50 - 100 mm) drilled or pushed into a slope or a dam, 30 - 100 m long with 2 - 5 % upward inclination, to drain the ground water from the slope face by gravity; the end is connected to a drain pipe. Ideal for landslides, cuttings, dam abutments.
- Spacing 3 - 15 m; the length extends beyond the potential slip surface. Pipes are slotted PVC wrapped with geotextile. Advantages: no pumping, low maintenance, lowers pore water pressure (r_u) and increases the factor of safety.

Foundation drains

- Foundation drain (perimeter drain / toe drain) is a trench around the foundation of a building or a retaining wall, below the footing level, with a perforated pipe (100 - 150 mm) in gravel envelope wrapped in filter fabric and sloping at 1 in 100 to a sump or storm drain.
- Purpose: remove the water around footings and basement, reduce the hydrostatic uplift and the lateral pressure on the walls, prevent the dampness and the softening and swelling of the soil.
- Under-drains below the floor slab (gravel + pipe) in basements and also behind the retaining wall (weep holes 75 mm at 3 m, with the gravel pocket).

Blanket drains

- Blanket (horizontal) drain is a layer of coarse granular material (sand and gravel 0.3 - 1.0 m thick, filter graded) laid under an embankment, earth dam, road, or a structure, to collect the seepage and discharge it to the toe drain.
- In earth dams it is the horizontal drain at the downstream third: keeps the phreatic line within the section, protects the downstream slope from the seepage and piping; the chimney drain is the vertical equivalent.
- Under road embankments it is a sub-base drainage layer (granular sub-base drainage layer, geotextile separator) to remove the water from the pavement and prevent the pumping and frost.
- Under buildings on expansive soil / floors: sand blanket (0.15 - 0.3 m) with perforated pipes.

Seepage water → Filter layer → Blanket / pipe → Toe drain or sump

Design of blanket drains (concept)

- Capacity: $q = k_d i t$ (k_d = permeability of the drain material, t = thickness).
- The drain permeability should be at least 100 times that of the soil being drained; adequate thickness; free from clogging by the filter layer.

5. Criteria for Selection of Fill Material Around Drains

The filter (envelope) material around a drain must (i) retain the soil particles so that no fines are washed in (piping) and (ii) allow water to pass freely (permeable), and (iii) not clog with time. These requirements are expressed by the Terzaghi filter criteria, extended by Bertram and the US Army Corps (USBR).

Filter criteria (Terzaghi - Bertram)

- Piping (retention) criterion: D_{15} (filter) / D_{85} (soil) < 5
- Also D_{50} (filter) / D_{50} (soil) < 25 and for uniform soils $D_{15f} / D_{85s} < 5$
- Permeability (drainage) criterion: D_{15} (filter) / D_{15} (soil) > 5
- Uniformity: filter gradation curve nearly parallel to that of the soil; coefficient of uniformity $C_u = D_{60}/D_{10} < 20$ for the filter.
- Fines: the filter should have less than 5 percent finer than 75 micron (No. 200 sieve), and no plastic fines.

- Pipe slot / hole criterion: D_{85} (filter) > 1.2 x the slot width or > 1.0 - 2 x the hole diameter (so that the filter does not enter the pipe).
- When the drain / filter is two stage, apply the criteria to each successive layer (soil -> fine filter -> coarse filter -> pipe).
- The filter must be durable (hard, round or sub-angular, no soft particles), not prone to chemical clogging, easy to compact without segregation.
- Geotextile filter may replace the sand filter: apparent opening size (AOS) $O_{95} < 2 - 3\ D_{85}$ of the soil; permittivity higher than needed; adequate clogging resistance (gradient ratio test).

Soil to drain	Typical filter / fill material
Clay / silt	Coarse sand (0.1 - 5 mm), concrete sand; geotextile wrapped gravel
Fine sand	Medium sand / fine gravel in 2 layers
Coarse sand and gravel	Gravel 5 - 20 mm; stone, pea gravel
Collection pipe	Gravel 10 - 40 mm with D_{85} larger than slots

Example

- Soil: D_{85} = 0.2 mm, D_{15} = 0.02 mm. Then the filter D_{15} must be < 5 x 0.2 = 1.0 mm and > 5 x 0.02 = 0.1 mm => choose D_{15} about 0.3 - 0.8 mm (medium sand).

6. Electro-osmosis

Electro-osmosis is the movement of pore water through a fine grained soil (silt, clay) from the anode (+) towards the cathode (-) when a direct current is passed through the soil. Used when the soil is too fine ($k < 10^{-5}$ cm/s) for the gravity or vacuum drainage; first used by Casagrande (1939) in Germany.

- Principle: clay particles are negatively charged and the water has a diffuse layer of positive ions (cations). Under the electric field the cations move to the cathode, dragging the water with them. The water collects at the cathode (a perforated well point) and is pumped away.
- Electrodes: anode - steel rod / pipe (metal, corrodes) or any, cathode - perforated well point (steel tube). Spacing 3 - 6 m (anode-cathode), voltage 40 - 180 V (gradient of 50 V/m), DC power from a generator.

Quantity of flow

- $q = k_e i_e A$ (k_e = coefficient of electro-osmotic permeability, i_e = voltage gradient (V/m), A = area)
- k_e is about 5 x 10^{-5} cm/s per V/cm and is almost independent of the grain size - so it is the same for all soil types (the reason it works well in clays: for clay $k_{hydraulic}$ about 10^{-8} but k_e about 5 x 10^{-5}).
- Example: i_e = 0.5 V/cm => electro-osmotic flow velocity v = 5 x 10^{-5} x 0.5 = 2.5 x 10^{-5} cm/s (about 2 cm/day).

Applications and effects

- Dewatering and stabilising silts and soft clays in deep excavations, slopes (stabilising of landslides, Norwegian clay), consolidation of soft clay, pile driving (reduces the friction, increases the strength after the current: electro-osmotic piles).
- Drainage of tunnels, increase in the shear strength by consolidation and by the chemical effect (electro-chemical hardening with injection of $CaCl_2$ or Al salts at the anode).
- Reduce the water content, increases the strength by 2 - 3 times. Also remediation of contaminated clays (electrokinetic).

Advantages	Limitations
Works in clays and silts where other methods fail	High power cost (energy about 20 - 200 kWh/m^3)
Quick effect on the strength; no effect on the nearby structures	Anode corrosion; the soil near the anode dries and cracks, resistance increases
Can be combined with the well points and sheet piles	Not suitable in salty soil (high conductivity - wasted current) or with buried metal

7. Grouting - Objectives and Grouts

Grouting is the process of injecting a fluid material (grout) under pressure into the voids, cracks or fissures of soil or rock, where it hardens (sets) and thereby improves the strength and reduces permeability of the ground.

Objectives (uses) of grouting

- Reduce permeability - seepage control (curtain grouting below dams, cut-off), tunnels, shafts, excavations.
- Increase strength and bearing capacity - strengthen weak soil and rock; underpin old foundations.
- Reduce settlement, fill voids (cavities, mine workings, sinkholes), compaction grouting.
- Stabilise loose sands against liquefaction, excavation stability, support of tunnels, seal leaking walls and rock.
- Fill the gap behind tunnel linings, pre-stress and anchor (rock bolt grouting), repair the concrete (crack repair), raising of slabs (slab jacking).

Grouts - types and properties

Grout	Composition	Use / remarks
Cement grout (suspension)	Cement + water (w/c 0.4 - 2.0), + bentonite 1 - 3 %, fly ash, admixtures	Rock fissures, coarse gravel and sand; cheap; particles > 0.1 mm - cannot enter fine sand (voids < 3 x D_{85})
Micro-fine cement	Cement < 10 micron (d_{95} < 15 micron), slag	Fine sand, narrow fissures
Clay (bentonite) grout	Bentonite + water; + cement	Cheap, low strength; seepage cut-off; reduces cement bleeding
Chemical: sodium silicate (water glass)	Sodium silicate + reagent ($CaCl_2$, formamide, bicarbonate) forms silica gel	Medium to fine sand; Joosten two shot, one shot Siroc; low viscosity; strength 0.5 - 3 MPa
Chemical: acrylamide, polyurethane, resins (epoxy)	Polymer grouts	Very fine sand; leak sealing; costly, toxicity (acrylamide banned in many countries)
Bituminous grout	Asphalt emulsion	Seepage sealing in rock
Lime / lime-fly ash, mortar grout	Lime slurry, sand-cement mortar	Large voids, cavities, mines; fill (compaction grout = low slump mortar)

Properties of the grouts

- Viscosity (resistance to flow - low viscosity enters finer pores), fluidity; Penetrability (the ability to enter the soil - groutability ratio $N = D_{15}$ (soil) / D_{85} (grout) > 25, and D_{10} (soil) / D_{95} (grout) > 11 for suspension grout).
- Setting time (gel time) - must be long enough to place and short enough to avoid the dilution; bleeding (water separation, < 2 - 5 % in 2 h), stability (no segregation), strength, durability, permeability after setting, shrinkage, compatibility with the ground water chemistry (sulphates), toxicity and cost.
- Rheology: Newtonian (chemical) and Bingham fluid (cement, clay - yield stress). Pressure and the rate of injection control the spread.
- Groutability: ratio of grain size of the soil to that of the grout. Particulate grout: soil permeability k > 10^{-1} cm/s; chemical grout k > 10^{-3} cm/s.

8. Grouting Methods - Ascending, Descending, Stage

In rock and in deep soil the grouting is done in sections (stages) of the hole length - usually 3 to 6 m each - using packers to isolate the section under treatment. The different sequences are called the

grouting methods.

1. Ascending (upstage) method

- The hole is drilled to the full depth at once. Grouting is begun from the bottom section and then the packer is raised to the next section above. (also called "packer method up-stage").
- Packer set above the section, grout is injected under the pressure, then raised to grout the next, upwards.
- Advantages: quick, drilling is done once, cheap; Disadvantages: the grout may leak (by-pass) along the hole and up, fissures upstream may be left ungrouted, pressure at the upper zone limited, not suitable for the badly fractured rock, the hole may collapse.

2. Descending (downstage) method

- The hole is drilled for the first stage (top section), grouted, the grout is allowed to set, the hole is re-drilled (washed) through the set grout and deeper by the next stage, and so on downwards.
- Advantages: the upper zone already sealed so that higher pressures may be used in the lower stages; best control; grouting of the poor rock is effective; less lifting and surface leaks; used for dam foundations, fractured rock and tunnels.
- Disadvantages: slow and costly (the waiting time for the setting, re-drilling).

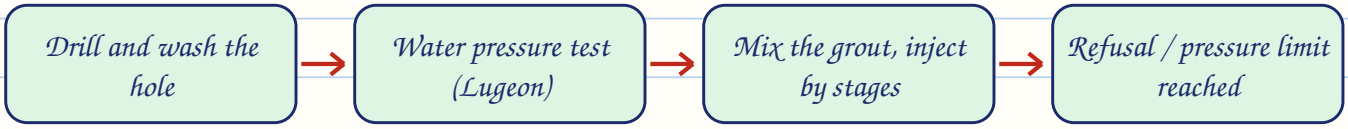
3. Stage grouting (Stage by stage / packer)

- The hole is divided into stages; each stage is grouted with a packer (single or double) and the sequence depends on the rock: stage grouting can be ascending stage or descending stage; the packer type is a single packer or double (straddle) packer to isolate a single zone.
- The stage length 3 - 6 m (3 m for poor rock, 6 m for good rock).
- Tube-a-manchette (sleeve pipe) method for soils: a plastic pipe with rubber sleeves at 0.33 - 1 m spacing is grouted into the bore with a sheath grout, then a double packer injects the grout through any sleeve at the needed pressure, multiple times - very controlled in alluvium and sand.

Aspect	Ascending	Descending
Drilling	Full depth once	Stage by stage, re-drilling
Time and cost	Less	More
Control and quality	Lower	Higher
Pressure	Limited by leakage	Higher in the lower stages
Best for	Good, sound rock; soil	Weak, broken and fissured rock, dams

Sequence of the holes (split spacing)

- Primary holes at 12 m spacing; secondary at 6 m; tertiary at 3 m (split spacing); check holes at the end. The grout take decreases in each series - the sign of a successful treatment.
- Curtain grouting (1 - 3 rows), blanket / consolidation grouting (shallow, closely spaced to the grid of 3 - 6 m), contact grouting (between the concrete and rock), jet grouting (high pressure 20 - 40 MPa jet cuts the soil and mixes the cement to create columns - soil-crete), compaction grouting (stiff mortar bulbs densify the soil), permeation grouting.



- Pressure limit: about 1 kPa per cm depth, i.e., $p = 0.23 - 0.25\ kg/cm^2$ per m depth of the overburden (about 22 kPa/m); to avoid heave and hydraulic fracture. Refusal when the take is < 1 l/min at the max pressure for 10 min.
- Grout mix changed: start thin (w/c = 5 : 1) and thicken (3:1, 2:1, 1:1, 0.8:1) as the take continues.

9. Hydraulic Fracturing in Soils and Rocks

Hydraulic fracturing (hydrofracture) is the splitting of the ground by the fluid pressure when the injection pressure exceeds the minimum principal stress plus the tensile strength of the soil / rock. The fracture is a new crack opened by the fluid, which is then filled by the grout - forming lenses and sheets.

Mechanism

- In a borehole the fluid pressure p_f is raised; around the hole the tangential (hoop) stress becomes tensile. When p_f reaches the breakdown pressure the soil fractures.
- Fracture occurs perpendicular to the minimum principal stress - in the field generally vertical fractures at the shallow depth where horizontal stress ($sigma_3$) is the minimum, and horizontal fractures where the vertical stress is the minimum (below about 8 - 10 m in the normally consolidated soil it is horizontal when the $K_0 > 1$).
- Breakdown pressure: $p_b = sigma_3 + T_0 - u$ (fracture in rock); in soil $p_b = 2\ sigma_3 - u + c$ (approx. from the cavity theory) - the typical injection pressure ratio $p_f / sigma_v$ about 1.2 - 2.0.

Uses

- Deliberate hydrofracture: to grout fine grained soils (clay, silt) that cannot be permeated - a network of grout sheets / lenses compresses the soil, raises the structure (compensation grouting

under the tunnel and buildings), soil improvement of clay, and to increase the permeability of the rock for oil and geothermal wells and groundwater wells (yield).

- Measurement of in-situ stress of rock (hydraulic fracturing test) - shut in pressure = sigma $_3$.
- Unwanted hydrofracture in dams: the cracks in the core of the earth dam (hydraulic fracturing of the core - a cause of dam piping), and in the grouting at too high pressure causing the heave or uplift of the ground and rock mass, opening the existing joints, and leaks.

In soils	In rocks
Fracture by the tension around the hole; grout lenses in the weakest plane; fine soils are strengthened by the grout lenses and consolidation	Opens the existing joints / bedding planes; the pressure limited by overburden (rock cover) to avoid the lifting
Pressure monitored by the pressure-time and flow-time curves - a sudden drop in pressure with an increase in flow indicates fracture	Pressure increased in steps; Lugeon test to check the permeability before grouting
Used in sleeve pipe in clay and compensation grouting	Used for the curtain below the dam, for gas, oil and geothermal wells

10. Post Grout Tests

After the grouting, quality control tests verify that the planned improvement (strength, permeability, continuity) has been achieved. They are done after the grout has cured (7 - 28 days).

Test	Purpose / procedure
Water pressure (Lugeon) test	Packer test in the check holes: water is injected at 10 kg/cm^2 for 10 min; 1 Lugeon = 1 litre/min per metre of the hole at 10 kg/cm^2; target < 1 - 3 Lugeon after the grouting
Permeability / pumping test	Insitu permeability with falling or constant head test in boreholes or pumping from wells
Core drilling and inspection	Check holes - cores show the grout in the fissures, Rock Quality Designation (RQD) improved
Unconfined compression test on the cores / samples	Strength of grouted soil; strength gain ratio
Standard penetration (SPT), cone penetration (CPT), pressure-meter	Compare the penetration resistance before and after (e.g., in sands N increases from 8 to 25)
Plate load test	Bearing capacity and the modulus after the grouting
Geophysical tests	Seismic cross-hole / down-hole velocity increases; electrical resistivity; ground penetrating radar
Excavation (pit) / visual	Directly see the grouted mass
Flow meter / piezometers	Seepage and the pore pressure under the dam compared with the design
Grout take records	Decreasing take from primary to tertiary holes proves the closure

- Check holes are drilled at 5 - 10 % of the total holes at the locations of the likely weakness (between the grout holes).
- The acceptance criteria are stated: (i) permeability less than the specified (say 1 x 10^{-5} cm/s, or 3 Lugeon), (ii) minimum strength, (iii) no loss of the water in the check holes.

Quick revision of Unit II

- Dewatering: sump (gravel, coarse sand), well points (sand; single < 6 m, multi stage deeper, vacuum for silty sand), horizontal drains, blanket drains, electro-osmosis for clay (k_e = 5 x 10^{-5} cm/s per V/cm).
- Filter: $D_{15f}/D_{85s} < 5$ and $D_{15f}/D_{15s} > 5$.
- Grout: cement, clay, chemical (silicate); groutability ratio; ascending / descending / stage (packer); hydrofracture when $p > sigma_3 + T$; post grout: Lugeon test, cores, SPT.

Important Questions (JNTUA pattern)

- Explain the methods of dewatering and the criteria for their selection with respect to the soil type.
- Write about sumps and interceptor ditches. What are their limitations?
- Explain the single stage, multi stage and vacuum well point systems with neat sketches.
- Explain the horizontal wells, foundation drains and blanket drains.
- What are the criteria for the selection of the filter material around drains? Give an example.
- Explain the principle and application of electro-osmosis. Derive the flow equation.
- What are the objectives of grouting? Explain the types of grouts and their properties.
- Explain ascending, descending and stage grouting methods and compare them.
- Write short notes on hydraulic fracturing in soils and rocks and post grout tests.